



**SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA**

CPC357: IOT ARCHITECTURE AND SMART APPLICATIONS

Project: Green House Monitoring with Automated Ventilation System

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Introduction

This document is prepared for our IoT Project: ***Green House Monitoring with Automatic Ventilation*** that targets ***UN's Sustainable Development Goal 2: Zero Hunger***. This IoT Project is a solution designed to optimize agricultural practices and ensure efficient resource management. By integrating real-time monitoring systems and automated ventilation mechanisms, this prototype creates a controlled environment that supports optimal plant growth, reduces resource wastage, and increases crop yield.

This system utilizes sensors to monitor key environmental parameters such as temperature, humidity, and carbon emissions in the environment. Based on this data, an automated ventilation system dynamically adjusts airflow and temperature, creating ideal conditions for plant growth. This not only maximizes productivity but also reduces dependency on external inputs, such as excessive water usage and energy-intensive cooling systems.

Hardware Integration

This IoT Project is created by using several sensors such as: **DHT11** (for temperature and humidity), **Soil Moisture** (for reading moisture content in the soil), and **MQ2 Sensor** (for detecting harmful gases such as CO and flammable gases). Along with these sensors we have used actuators such as: **SG90 Servo Motor** (for opening ventilation window), and a **Shaft Motor** (for driving out the harmful gases and letting fresh air in). All these components were then connected to a micro-controller unit (**ESP32**) to make everything work.

We used jumper wires and a breadboard to make connections for all these components. We referred to the Cytron website for the documentation and programming guide for all of these different components.

Software Integration

We used **Arduino IDE** with **C++** language to program this system. We began by first importing all the required libraries that are **<DHT.h>**, **<ESP32Servo.h>**, and **<VOneMqttClient.h>**. We had to look at some documentation on the internet on how to use these libraries properly.

Cloud Integration

For this prototype, we have used **V-One IoT Platform** for uploading data from our IoT system to the cloud in real-time. Here is the screenshot of the Dashboard we created on the V-One IoT Platform.

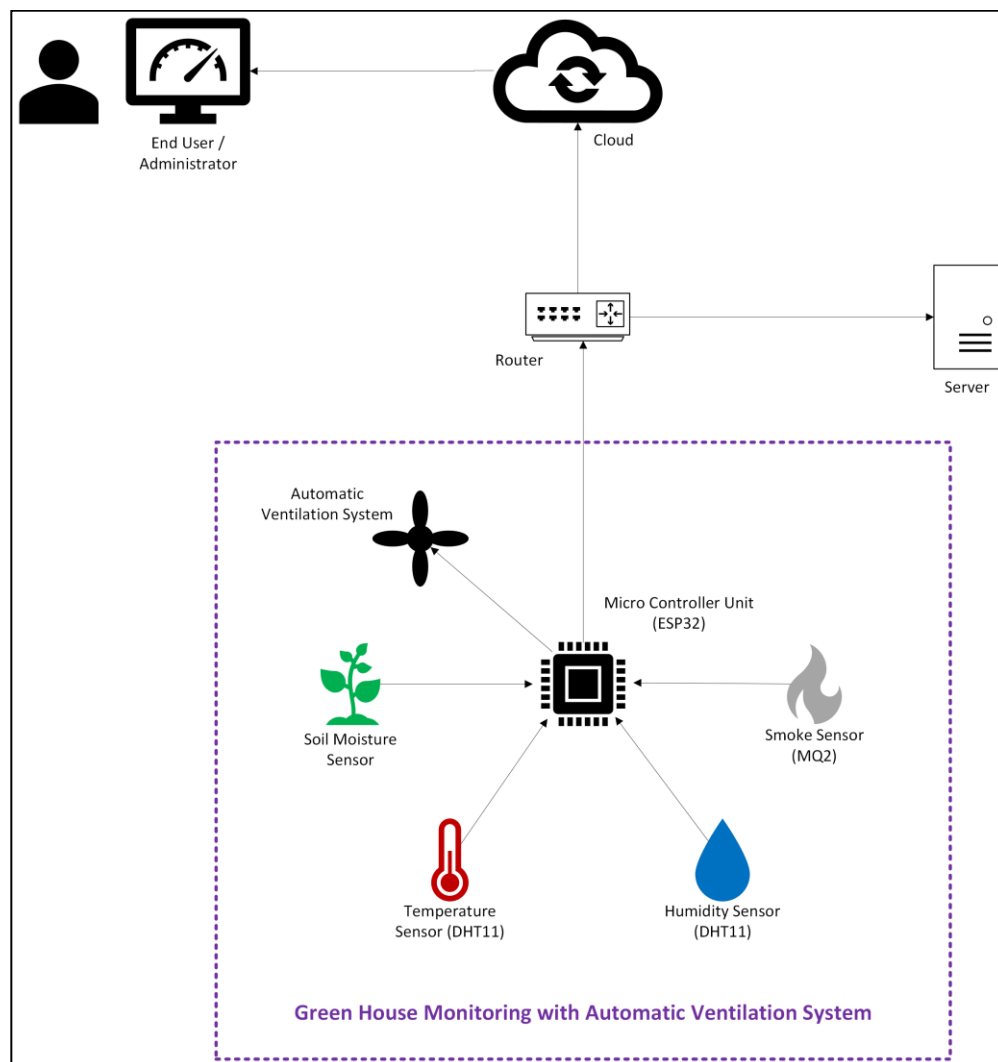


We have also color coded the graphs for a *user-friendly dashboard*

System Architecture and Setup Guide

System architecture for this project includes the Green House Monitoring system equipped with an automatic ventilation system, a soil moisture sensor, temperature and humidity sensor, a smoke detector sensor, and a micro-controller unit to interconnect all these components. The micro-controller streams data to cloud through a router connected to the internet and then the data from the cloud can be presented to the administrator through the Monitoring dashboard facilitated by the cloud platform.

Setup guide includes connecting the IoT Device with a computer with monitoring dashboard opened and powering up the IoT Device connected to the computer. The IoT Device will immediately start to stream data to V-One IoT Platform.



Source Code

GitHub Repository Link: [Green House Monitoring with Automatic Ventilation System](#)

Video Presentation

Video Presentation + Live Demo: [Click Here](#)